Jay Hemnani

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Gujarat, India (Open to Relocate)

Education

Pandit Deendayal Energy University (PDEU) 2018 – 2022 | Gandhinagar, India
B.Tech in Computer Engineering | GPA: 8.7/10

- **Coursework:** Computer Vision, Natural Language Processing, Big Data Analytics, Database Systems.
- **Publications:** 2 IEEE papers (AIMV 2021): Diabetes Prediction using Stacking Classifier, Efficient Vaccine Scheduler Based on CPU Scheduling Algorithms.

Technical Skills

Programming: Python, SQL, Java, C++, JavaScript/TypeScript
ML/AI Frameworks: PyTorch, PyTorch Geometric, TensorFlow, Scikit-learn, Hugging Face Transformers, YOLOv8
ML Engineering: FAISS, LangChain, LangGraph, Vector Databases (Pinecone, ChromaDB), RAG Pipelines, Feature Engineering
Data Processing: Pandas, NumPy, PySpark, Apache Kafka, Airflow
Cloud & MLOps: AWS (SageMaker, Lambda, S3), Docker, Kubernetes, MLflow, DVC, Weights & Biases
Databases: PostgreSQL, MongoDB, Redis, TimescaleDB, ChromaDB, Pinecone

Experience

Data Analyst, *Elite Hotel Group* Summer 2025 | Remote

- Built **time-series demand forecasting models** for dynamic pricing and revenue management decisions.
- Defined **metrics and SLAs** with finance and operations; they became the spec for the ETL and forecasting work.
- Built **automated ETL pipelines** in SQL and Python, cutting manual data preparation by **40%**.

Technical Consultant, *Independent* 2022 – 2024 | Remote

- **Data analytics and pipeline consulting** for small businesses: reporting automation, data infrastructure, A/B testing.

AI/ML Intern, *Amnex* Jan – May 2022 | Gujarat, India

- Built a **credit fraud detection ensemble** (Random Forest, XGBoost, SMOTE); **94% precision** on transaction data.
- Built **automated analytics dashboards** on statistical models for operational reporting and anomaly detection.

Software Engineering Intern, *Cactus Creatives* May – Nov 2019 | Gujarat, India

- Built a **cloud-native communication platform** on Azure; set up **CI/CD pipelines** that cut deployment time by **60%**.

Projects

Soccer Vision Research | *Python, PyTorch, RF-DETR, SAM2, SigLIP*

- Built a **modular CV research framework** over 4 pretrained models: RF-DETR detection, SAM2 segmentation and tracking, SigLIP zero-shot identification, ResNet jersey classification.
- 3 execution modes (sequential, parallel, adaptive) and a fusion layer with **7 selectable strategies**, using Hungarian matching to align detections across models.
- Schema-validated YAML config and a model registry, so swapping a model or preset is a config edit, not a code change.

Nobel Data Intelligence | *Python, PyTorch, ProDy, Transformers, RDKit, Biopython*

- Built a **tri-modal model** for protein stability and enzyme kinetics: ProtT5 for sequence, a CNN over a 1000-point **VDOS vibrational spectrum** (normal mode analysis via ProDy), and ChemBERTa for chemistry.
- A learned gate weights the three branches per prediction; MM-Drop training masks the spectral branch so the model still works when no structure is available.

FIFA Soccer DS | *Python, YOLOv8, ByteTrack, PyTorch Geometric, FastAPI, MLflow, DVC*

- YOLOv8n detection and ByteTrack tracking on La Liga footage at **22 FPS** on an RTX 3070 laptop. Tracklets feed a GraphSAGE role classifier that is wired in but not yet trained.
- FastAPI service with a live RTSP path, ONNX and TensorRT export, MLflow tracking and a DVC pipeline stage.

Open Source Contributions

220+ merged pull requests across 37 open-source projects, including vLLM, DSPy, the MCP Python SDK, Modular, Effect, Caddy, Lightpanda, radare2 and Home Assistant.
A plain-English note on each one: jayhemnani.in/oss